

MXCuBE site report

EMBL Hamburg

Brief beamlines summary

- P13 High throughput MX
 - ~~4-17 keV, 30 μ m beam, 5e12 ph/sec, 7 ms exposures~~
 - ~~Fixed 11.7 keV, 5e12 ph/sec, <300 μ m beam~~ *optics upgrade in progress*
 - ~~Fixed 12.7 keV, 1e13 ph/sec, <300 μ m beam~~ *DCM mechanical issues, optics upgrade in progress*
 - 6 - 18 keV, 6e12 ph/sec, <300 μ m **top-hat beam** - optics upgraded, under MXCuBE control
 - EIGER 16M Si, MD2
 - MARIVN sample change, SPINE 170 samples, 50 sec. sample cycle
- P14-1
 - High energy, ultrahigh resolution MX; micro focus MX; serial MX
 - HiTT - High Throughput Tomography, X-ray imaging
 - 7-32 keV, 1 μ m-1400 μ m beam, 2e13 ph/sec
 - EIGER2 16M CdTe, MD3 DOWN, MiniKappa, PlateScanner, Viscous Jet, On-axis lasers
 - MARVIN sample change, SPINE 170 samples, 50 sec. sample cycle
- P14-2 T-REXX, time-resolved MX
 - 12.7-**15** keV, 10x15 μ m beam, 3e12 ph/sec
 - EIGER 4M Si, ARINAX BSD, SmarAct stages
 - patterned chips, environment chamber 10-80 $^{\circ}$ C $\pm$ 2%RH, On-axis lasers, nano-drop injector
 - **NEW: EF-X - electric-field stimulated crystallography set up, 1 MV/cm, 50-500 ns pulse**
 - **NEW: Anaerobic sample chamber (patterned chips), < 100 ppm O₂**

MXCuBE status

- No developer since December 2021
- In production, stable:
 - Python 2.7
 - Qt4
 - HardwareRepository Master 03/2021
 - Ubuntu 22.04
 - No merge between P13, P14, TREXX, no repo sync since then
- Security
 - interactive remote logins with one-off passwords under supervision, outgoing connections only
 - incoming connections for data download only
 - no horizontal access propagation through EMBL
 - CrowdStrike accident: during the startup in Apr 2026 beamlines crippled for several days by installation of Falcon “security” module which is blocking thread prioritization

Developments since last meeting

- Global Phasing Workflows - GPhL
 - Continuous improvements in strategies and processing
 - **NEW: Latest update 10/04/2026, ASTRAWorkflows-2.2.1**
- Data management
 - MXCuBE registers each RAW data set (MX, TREXX, HiTT) in EMBL-Heidelberg DMA (Data Management Application, F. McAdam O'Connel, W. Murphy, M. Helmling, EMBL-HD)
 - **NEW: Pilot project on re-archiving all EMBL MX RAW data of 2019 at dma.embl.de with links to PDB**
 - MCXuBE handles each HiTT PROCESSED data set to the WebKnossos instance
 - (<http://webknossos.embl-hamburg.de>), triggers OME-Zarr conversion (A. Svetlove, EMBL-HH))
- Remote access to compute cluster (beamline) users for (re-) processing:
 - 2FA, keycloak (EMBL-HH IT)

Composite OAV image $\sim 12 \times 12 \text{ mm}^2$ in 30 sec

- Targeting large samples for X-RAY Imaging
- Scan@control system level (TINE middleware server, *loze*)
- Direct camera connection (A-zoom) latencies on MD ImageJPEG property
- initially by Romain Carlino (ARINAX)
- Marina Nikolova



Plans for the next six months

- Installation of Double Multilayer Monochromator at P14, 1-1.5% dE/E, $> 10^{15}$ ph/s
 - physical installation starts next week, commissioning in 2027
- Long shutdown Sep 2026 - April 2027
- Re-archiving *all* RAW_DATA at dma.embl.de
- GPhL workflows at P13
- If would get an MXcUBE developer (though we most likely won't...)
 - merge local P13, P14, P14-GPhL and T-REXX branches
 - Update to python3, Qt7
 - Update to mxcube-core

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